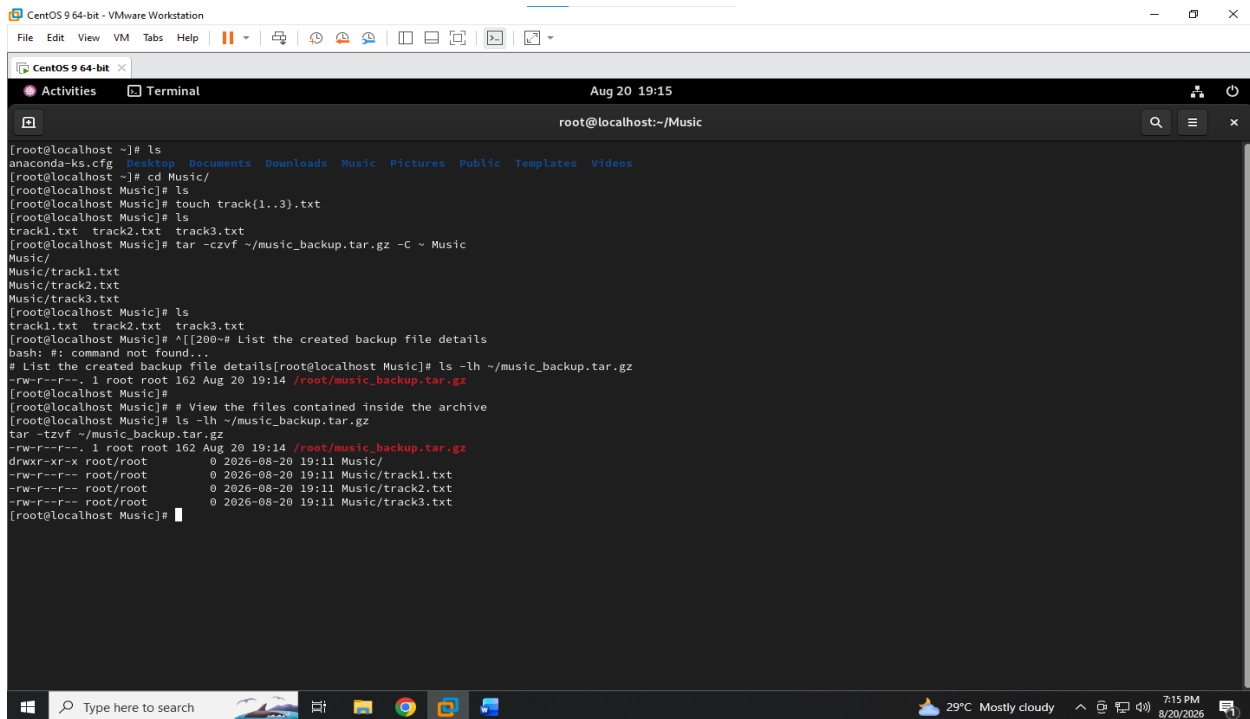


MODULE 8 - BACKUP AND RESTORE

GUIDED BY : VIPUL MISTRI

Q.1 Create a backup of your Music folder (or any folder with at least 3 files) using the tar command and compress it with gzip, saving the output as music_backup.tar.gz in your home directory.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Aug 20 19:15
root@localhost:~/Music

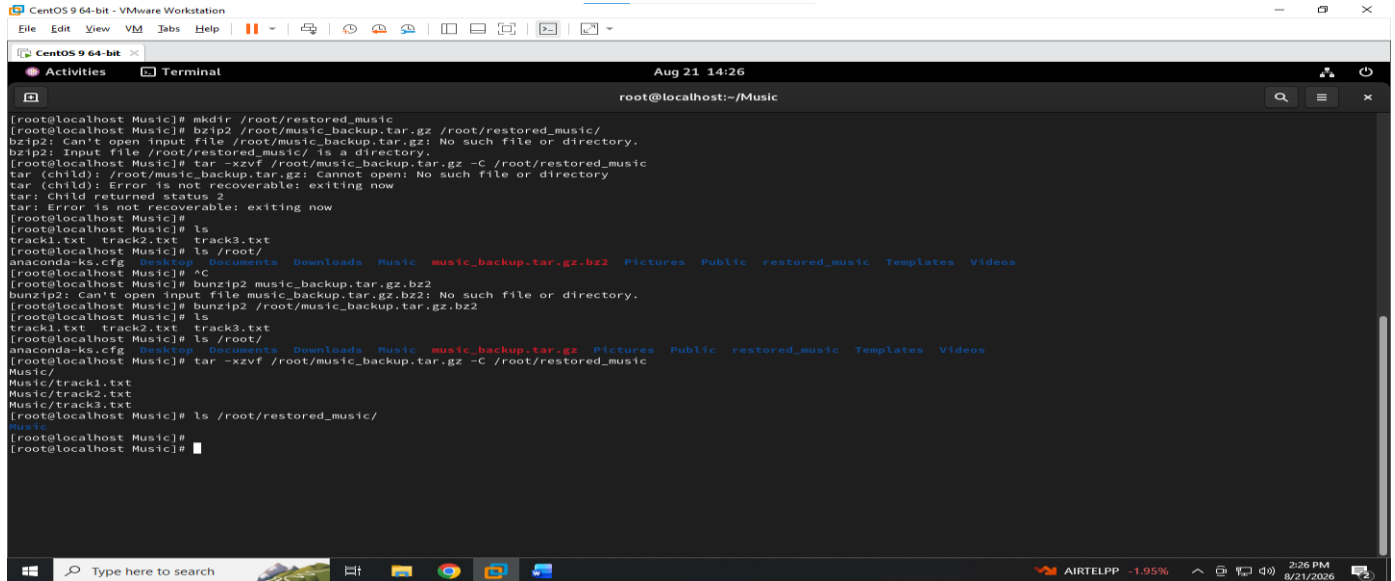
[root@localhost ~]# ls
anaconda-ks.cfg Desktop Documents Downloads Music Pictures Public Templates Videos
[root@localhost ~]# cd Music/
[root@localhost Music]# ls
[root@localhost Music]# touch track{1..3}.txt
[root@localhost Music]# ls
track1.txt track2.txt track3.txt
[root@localhost Music]# tar -czvf ~/music_backup.tar.gz -C ~ Music
Music/
Music/track1.txt
Music/track2.txt
Music/track3.txt
[root@localhost Music]# ls
track1.txt track2.txt track3.txt
[root@localhost Music]# ^[[200~# List the created backup file details
bash: #: command not found...
# List the created backup file details[root@localhost Music]# ls -lh ~/music_backup.tar.gz
-rw-r--r--. 1 root root 162 Aug 20 19:14 /root/music_backup.tar.gz
[root@localhost Music]#
[root@localhost Music]# # View the files contained inside the archive
[root@localhost Music]# ls -lh ~/music_backup.tar.gz
tar -tzvf ~/music_backup.tar.gz
-rw-r--r--. 1 root root 162 Aug 20 19:14 /root/music_backup.tar.gz
drwxr-xr-x root/root      0 2026-08-20 19:11 Music/
-rw-r--r-- root/root      0 2026-08-20 19:11 Music/track1.txt
-rw-r--r-- root/root      0 2026-08-20 19:11 Music/track2.txt
-rw-r--r-- root/root      0 2026-08-20 19:11 Music/track3.txt
[root@localhost Music]#
```

MODULE 8 - BACKUP AND RESTORE

GUIDED BY : VIPUL MISTRI

Q.2 Extract the contents of a tar.gz file (for example, music_backup.tar.gz) into a new directory called restored_music and verify that all files are restored correctly.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit
Activities Terminal
Aug 21 14:26
root@localhost: ~/Music

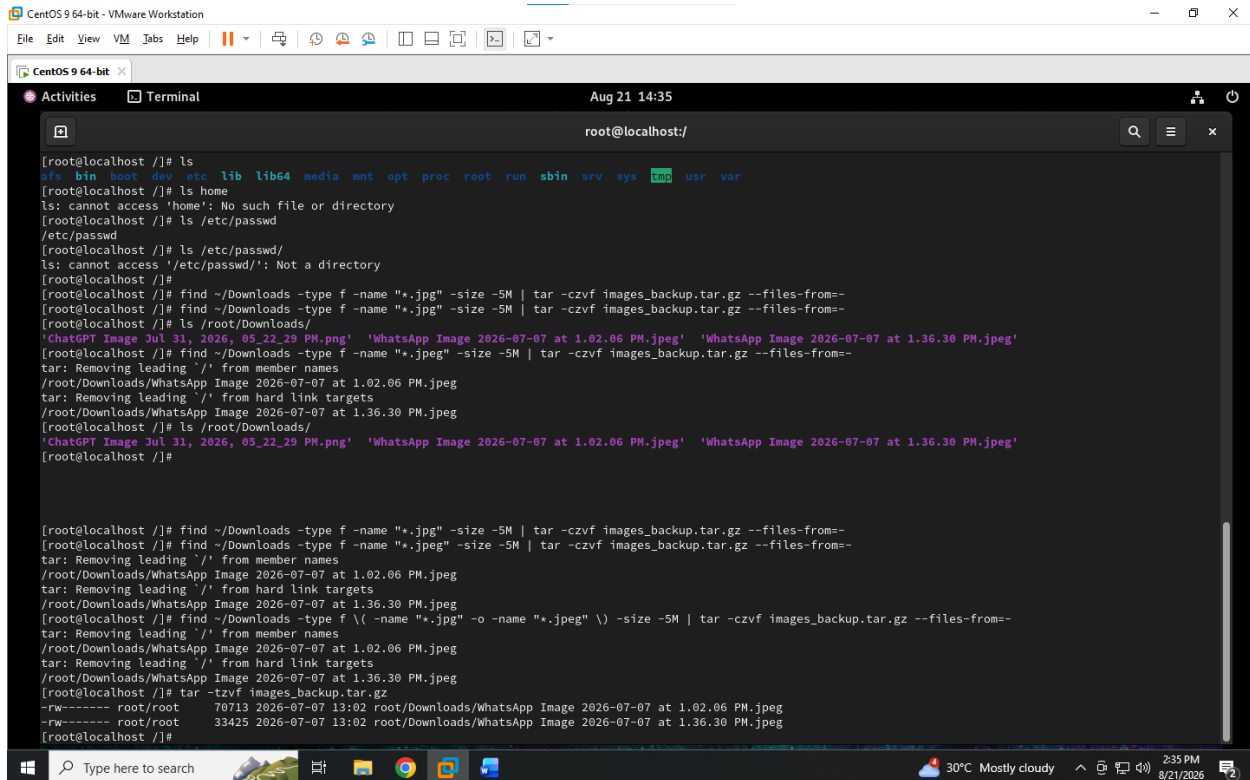
[root@localhost Music]# mkdir /root/restored_music
[root@localhost Music]# bzip2 /root/music_backup.tar.gz /root/restored_music/
bzip2: Can't open input file /root/music_backup.tar.gz: No such file or directory.
bzip2: Input file /root/restored_music/ is a directory.
[root@localhost Music]# tar -xzf /root/music_backup.tar.gz -C /root/restored_music
tar (child): /root/music_backup.tar.gz: Cannot open: No such file or directory
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
[root@localhost Music]#
[root@localhost Music]# ls
track1.txt track2.txt track3.txt
[root@localhost Music]# ls /root/
anaconda-ks.cfg Desktop Documents Downloads Music music_backup.tar.gz.bz2 Pictures Public restored_music Templates Videos
[root@localhost Music]# bunzip2 music_backup.tar.gz.bz2
bunzip2: Can't open input file music_backup.tar.gz.bz2: No such file or directory.
[root@localhost Music]# bunzip2 /root/music_backup.tar.gz.bz2
[root@localhost Music]# ls
track1.txt track2.txt track3.txt
[root@localhost Music]# ls /root/
anaconda-ks.cfg Desktop Documents Downloads Music music_backup.tar.gz Pictures Public restored_music Templates Videos
[root@localhost Music]# tar -xzf /root/music_backup.tar.gz -C /root/restored_music
Music/
Music/track1.txt
Music/track2.txt
Music/track3.txt
[root@localhost Music]# ls /root/restored_music/
Music
[root@localhost Music]#
[root@localhost Music]#
```

MODULE 8 - BACKUP AND RESTORE

GUIDED BY : VIPUL MISTRI

Q.3 Write a shell command that creates a compressed backup of all .jpg images from your Downloads folder into a file named images_backup.tar.gz, excluding any files larger than 5MB.
Hint: Use the find command with size filtering and tar with --files-from.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
Aug 21 14:35
root@localhost:/

[root@localhost /]# ls
afs bin boot dev etc lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
[root@localhost /]# ls home
ls: cannot access 'home': No such file or directory
[root@localhost /]# ls /etc/passwd
/etc/passwd
[root@localhost /]# ls /etc/passwd/
ls: cannot access '/etc/passwd/': Not a directory
[root@localhost /]#
[root@localhost /]# find ~/Downloads -type f -name "*.jpg" -size -5M | tar -czvf images_backup.tar.gz --files-from=
[root@localhost /]# find ~/Downloads -type f -name "*.jpeg" -size -5M | tar -czvf images_backup.tar.gz --files-from=
[root@localhost /]# ls /root/Downloads/
'ChatGPT Image Jul 31, 2026, 05_22_29 PM.png' 'WhatsApp Image 2026-07-07 at 1.02.06 PM.jpeg' 'WhatsApp Image 2026-07-07 at 1.36.30 PM.jpeg'
[root@localhost /]# find ~/Downloads -type f -name "*.jpeg" -size -5M | tar -czvf images_backup.tar.gz --files-from=
tar: Removing leading '/' from member names
/root/Downloads/WhatsApp Image 2026-07-07 at 1.02.06 PM.jpeg
tar: Removing leading '/' from hard link targets
/root/Downloads/WhatsApp Image 2026-07-07 at 1.36.30 PM.jpeg
[root@localhost /]# ls /root/Downloads/
'ChatGPT Image Jul 31, 2026, 05_22_29 PM.png' 'WhatsApp Image 2026-07-07 at 1.02.06 PM.jpeg' 'WhatsApp Image 2026-07-07 at 1.36.30 PM.jpeg'
[root@localhost /]#

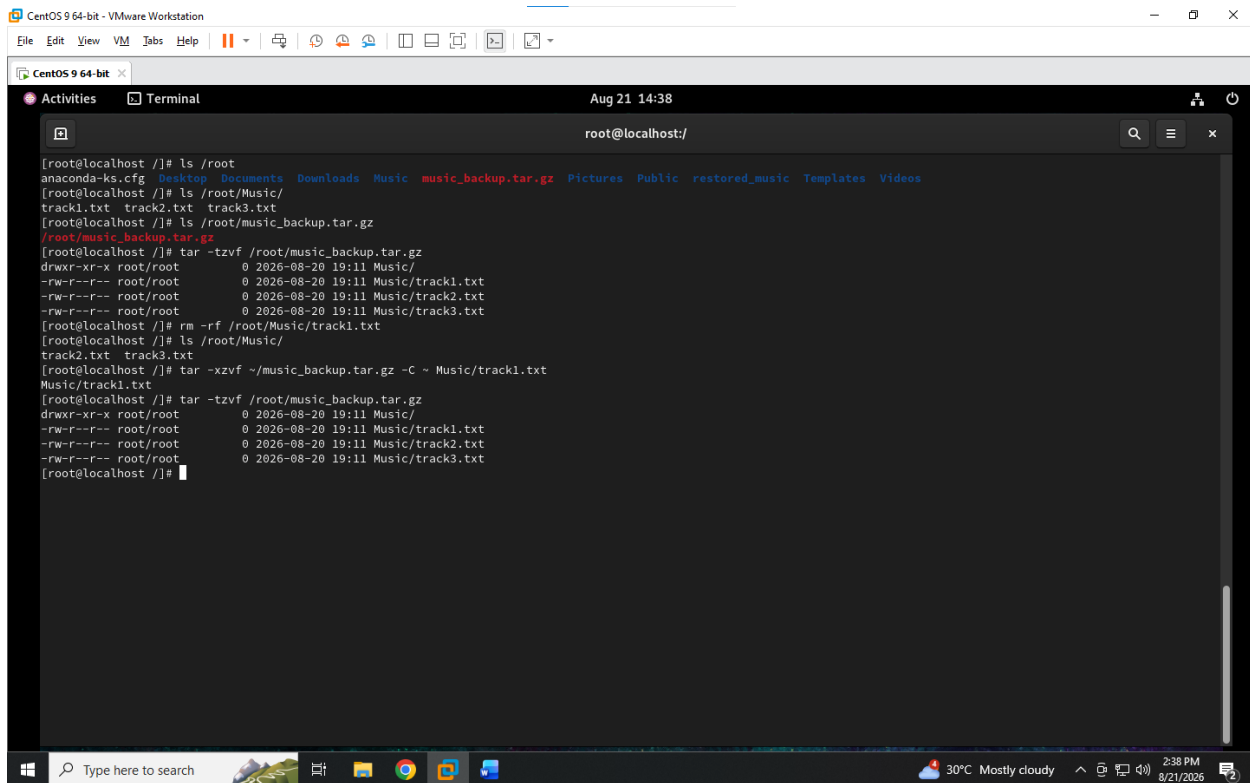
[root@localhost /]# find ~/Downloads -type f -name "*.jpg" -size -5M | tar -czvf images_backup.tar.gz --files-from=
[root@localhost /]# find ~/Downloads -type f -name "*.jpeg" -size -5M | tar -czvf images_backup.tar.gz --files-from=
tar: Removing leading '/' from member names
/root/Downloads/WhatsApp Image 2026-07-07 at 1.02.06 PM.jpeg
tar: Removing leading '/' from hard link targets
/root/Downloads/WhatsApp Image 2026-07-07 at 1.36.30 PM.jpeg
[root@localhost /]# find ~/Downloads -type f \( -name "*.jpg" -o -name "*.jpeg" \) -size -5M | tar -czvf images_backup.tar.gz --files-from=
tar: Removing leading '/' from member names
/root/Downloads/WhatsApp Image 2026-07-07 at 1.02.06 PM.jpeg
tar: Removing leading '/' from hard link targets
/root/Downloads/WhatsApp Image 2026-07-07 at 1.36.30 PM.jpeg
[root@localhost /]# tar -tzvf images_backup.tar.gz
-rw-r----- root/root      70713 2026-07-07 13:02 root/Downloads/WhatsApp Image 2026-07-07 at 1.02.06 PM.jpeg
-rw-r----- root/root      33425 2026-07-07 13:02 root/Downloads/WhatsApp Image 2026-07-07 at 1.36.30 PM.jpeg
[root@localhost /]#
```

MODULE 8 - BACKUP AND RESTORE

GUIDED BY : VIPUL MISTRI

Q.4 Simulate a restore scenario: delete one file from your original Music folder, then use your music_backup.tar.gz archive to restore only the deleted file without overwriting the rest.

ANS :



The screenshot shows a terminal window titled "CentOS 9 64-bit - VMware Workstation" with a menu bar (File, Edit, View, VM, Tabs, Help) and a toolbar. The terminal content is as follows:

```
root@localhost:/  
[root@localhost /]# ls /root  
anaconda-ks.cfg Desktop Documents Downloads Music music_backup.tar.gz Pictures Public restored_music Templates Videos  
[root@localhost /]# ls /root/Music/  
track1.txt track2.txt track3.txt  
[root@localhost /]# ls /root/music_backup.tar.gz  
/root/music_backup.tar.gz  
[root@localhost /]# tar -tzvf /root/music_backup.tar.gz  
drwxr-xr-x root/root          0 2026-08-20 19:11 Music/  
-rw-r--r-- root/root          0 2026-08-20 19:11 Music/track1.txt  
-rw-r--r-- root/root          0 2026-08-20 19:11 Music/track2.txt  
-rw-r--r-- root/root          0 2026-08-20 19:11 Music/track3.txt  
[root@localhost /]# rm -rf /root/Music/track1.txt  
[root@localhost /]# ls /root/Music/  
track2.txt track3.txt  
[root@localhost /]# tar -xzvf ~/music_backup.tar.gz -C ~ Music/track1.txt  
Music/track1.txt  
[root@localhost /]# tar -tzvf /root/music_backup.tar.gz  
drwxr-xr-x root/root          0 2026-08-20 19:11 Music/  
-rw-r--r-- root/root          0 2026-08-20 19:11 Music/track1.txt  
-rw-r--r-- root/root          0 2026-08-20 19:11 Music/track2.txt  
-rw-r--r-- root/root          0 2026-08-20 19:11 Music/track3.txt  
[root@localhost /]#
```

The terminal window is part of a desktop environment with a taskbar at the bottom showing a search bar, application icons, and system status (30°C, Mostly cloudy, 2:38 PM, 8/21/2026).